



Project Proposal: Mamba Biometric EKG Analysis Technology (MambaBEAT)

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1 Introduction

The field of Electrocardiogram (EKG/ECG) analysis has seen significant advancements with the advent of deep learning models. State-of-the-art models such as ResNet, Transformer, ViT, and H3 have demonstrated remarkable performance in EKG classification tasks. However, the quest for more efficient and accurate models is a continuous journey. This proposal introduces a novel project, provisionally named Mamba Biometric EKG Analysis Technology (MambaBEAT), which aims to leverage the Mamba model, a recent and promising sequence modeling architecture, for EKG analysis.

2 Literature review and Key Insights

The Mamba model, introduced in the paper “Mamba: Linear-Time Sequence Modeling with Selective State Spaces” [2312.00752], presents a unique approach to sequence modeling. It addresses the limitations of transformer models, particularly in processing long sequences, by combining linear-time sequence modeling with selective state spaces. The model’s efficiency is further enhanced by a hardware-aware parallel algorithm and a content-based propagation or forgetting strategy.

The Mamba model has shown state-of-the-art performance across several modalities such as language, audio, and genomics. However, its application in the field of EKG analysis remains unexplored. This presents an exciting opportunity for our project, MambaBEAT, to pioneer the use of the Mamba model in EKG analysis and classification.

Our literature review also included papers on HiPPO: Recurrent Memory with Optimal Polynomial Projections [2], How to Train Your HiPPO: State Space Models with Generalized Orthogonal Basis Projections [3], and Combining Recurrent, Convolutional, and Continuous-time Models with Linear State-Space Layers [4]. These papers provided valuable insights into state space models, orthogonal basis projections, and the combination of different types of models, which could be beneficial in our project.

3 Project Milestones

Our project plan involves exploring the Mamba model for EKG analysis. We will use publicly available EKG datasets for training and testing our model. The model’s performance will be evaluated based on its accuracy in classifying different EKG signals. We aim to outperform existing state-of-the-art models in terms of both accuracy and computational efficiency.

3.1 April 12th

Complete the data collection and preprocessing stage. Establish a baseline model for EKG classification using existing state-of-the-art models.

3.2 April 19th

Implement the Mamba model for EKG data and compare its performance with the baseline model.

4 Project Plan

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In conclusion, the MambaBEAT project represents a novel and ambitious endeavor in the field of EKG analysis. By leveraging the Mamba model, we hope to push the boundaries of what is currently achievable in EKG classification tasks.

References

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